The d- and f-Block Elements: Master Tutorial & Question Bank

Complete conceptual coverage of 3d-series transition metals, lanthanoids, actinoids, coordination redox trends, potassium dichromate, and potassium permanganate with exam-aligned practice.

1. Overview & Learning Objectives

The *d*-block and *f*-block elements encompass the vast majority of all metallic elements in the periodic table. For the NEET UG examination, mastery over electronic configurations, abnormal trends in ionization enthalpies, variable oxidation states, redox electrode potentials (E°), lanthanoid contraction, and the preparation and stoichiometric reactions of potassium dichromate ($\mathrm{K_2Cr_2O_7}$) and potassium permanganate ($\mathrm{KMnO_4}$) is imperative.

CORE NEET LEARNING GOALS

- **Position & IUPAC Definition:** Understand why Group 12 elements (Zn, Cd, Hg) are classified as *d*-block elements but *not* as transition elements.
- **3d Series Periodic Anomalies:** Explain the half-filled and completely filled configurations of Cr and Cu, high enthalpies of atomization, and irregularities in first, second, and third ionization enthalpies.
- **Redox Chemistry & Standard Potentials:** Rationalize why Cu has a positive $E^\circ(\mathrm{M^{2+}/M})$ value, why $\mathrm{Cr^{2+}}$ is a reducing agent while $\mathrm{Mn^{3+}}$ is a powerful oxidizing agent, and why $\mathrm{Cu^+(aq)}$ disproportionates.
- **Compounds of Chromium and Manganese:** Master the commercial and laboratory preparations, acid-base interconversions, tetrahedral geometries, and quantitative redox reactions of $\mathrm{K_2Cr_2O_7}$ and $\mathrm{KMnO_4}$.
- **Inner Transition Metals (f-Block):** Contrast lanthanoids with actinoids in terms of oxidation states, magnetic complexity, chemical reactivity, and the origin and consequences of lanthanoid and actinoid contractions.

2. IUPAC Definition & Electronic Configurations

2.1 Transition Metals vs. Group 12 Elements

According to IUPAC, a **transition element** is defined as an element having an incompletely filled (partially filled) *d*-subshell in its neutral atom or in any of its common oxidation states.

- **Scandium ($Z = 21$):** Has ground state configuration $[\mathrm{Ar}]\,3d^1 4s^2$. Because its neutral atom possesses a partially filled $3d$ orbital ($3d^1$), it is a transition metal.
- **Zinc ($Z = 30$), Cadmium ($Z = 48$), and Mercury ($Z = 80$):** Possess full $(n-1)d^{10}$ configurations in their elemental ground states ($d^{10}s^2$) as well as in their common oxidation states ($\mathrm{Zn^{2+}}$, $\mathrm{Cd^{2+}}$, $\mathrm{Hg^{2+}}$ are all d^{10}). Hence, they are **not** regarded as transition metals, though they are studied with the *d*-block because they represent the terminal members of each series.
- **Silver ($Z = 47$):** Ground state configuration is $[\mathrm{Kr}]\,4d^{10}5s^1$. While its neutral atom has a full $4d$ subshell, silver exhibits a $+2$ oxidation state in compounds like $\mathrm{AgF_2}$, where it possesses a partially filled $4d^9$ configuration. Thus, silver is definitively classified as a transition element.

2.2 General Electronic Configuration

The general valence electronic configuration of the *d*-block is:

GENERAL VALENCE SHELL CONFIGURATION

$$(n-1)d^{1-10}\,ns^{0-2}$$

Note: Palladium (Pd, $Z = 46$) has the outer configuration $4d^{10}5s^{0}$, completely emptying its outer s-orbital.

Table 1: Ground-State Electronic Configurations of the First Row (3d) Transition Elements

Element	Symbol	Z	Neutral Atom (M)	M^{+}	M^{2+}	M^{3+}
Scandium	Sc	21	$[\mathrm{Ar}]\,3d^{1}4s^{2}$	$[\mathrm{Ar}]\,3d^{1}4s^{1}$	$[\mathrm{Ar}]\,3d^{1}$	$[\mathrm{Ar}]\,3d^{0}$
Titanium	Ti	22	$[\mathrm{Ar}]\,3d^{2}4s^{2}$	$[\mathrm{Ar}]\,3d^{2}4s^{1}$	$[\mathrm{Ar}]\,3d^{2}$	$[\mathrm{Ar}]\,3d^{1}$
Vanadium	V	23	$[\mathrm{Ar}]\,3d^{3}4s^{2}$	$[\mathrm{Ar}]\,3d^{3}4s^{1}$	$[\mathrm{Ar}]\,3d^{3}$	$[\mathrm{Ar}]\,3d^{2}$
Chromium	Cr	24	$[\mathrm{Ar}]\,3d^{5}4s^{1}$	$[\mathrm{Ar}]\,3d^{5}$	$[\mathrm{Ar}]\,3d^{4}$	$[\mathrm{Ar}]\,3d^{3}$
Manganese	Mn	25	$[\mathrm{Ar}]\,3d^{5}4s^{2}$	$[\mathrm{Ar}]\,3d^{5}4s^{1}$	$[\mathrm{Ar}]\,3d^{5}$	$[\mathrm{Ar}]\,3d^{4}$
Iron	Fe	26	$[\mathrm{Ar}]\,3d^{6}4s^{2}$	$[\mathrm{Ar}]\,3d^{6}4s^{1}$	$[\mathrm{Ar}]\,3d^{6}$	$[\mathrm{Ar}]\,3d^{5}$
Cobalt	Co	27	$[\mathrm{Ar}]\,3d^{7}4s^{2}$	$[\mathrm{Ar}]\,3d^{7}4s^{1}$	$[\mathrm{Ar}]\,3d^{7}$	$[\mathrm{Ar}]\,3d^{6}$
Nickel	Ni	28	$[\mathrm{Ar}]\,3d^{8}4s^{2}$	$[\mathrm{Ar}]\,3d^{8}4s^{1}$	$[\mathrm{Ar}]\,3d^{8}$	$[\mathrm{Ar}]\,3d^{7}$
Copper	Cu	29	$[\mathrm{Ar}]\,3d^{10}4s^{1}$	$[\mathrm{Ar}]\,3d^{10}$	$[\mathrm{Ar}]\,3d^{9}$	–
Zinc	Zn	30	$[\mathrm{Ar}]\,3d^{10}4s^{2}$	$[\mathrm{Ar}]\,3d^{10}4s^{1}$	$[\mathrm{Ar}]\,3d^{10}$	–

NEET RULE: ORDER OF ELECTRON LOSS

When transition elements ionize, **outer $4s$ electrons are always removed before inner $3d$ electrons**. Therefore, all divalent ions (M^{2+}) of the $3d$ series possess exclusively $3d^{n}$ configurations with no valence $4s$ electrons.

3. General Physical Trends in the 3d Transition Series

3.1 Atomic and Ionic Radii

Across the $3d$ series from Sc to Zn, atomic radii show a characteristic non-linear profile:

- **Initial Decrease ($\mathrm{Sc} \rightarrow \mathrm{Cr}$):** Nuclear charge increases by 1 at each step, and electrons enter inner $3d$ orbitals. The shielding provided by d electrons is imperfect, so effective nuclear charge (Z_{eff}) increases, pulling the outer electron cloud inward.
- **Nearly Constant Mid-Series ($\mathrm{Fe}, \mathrm{Co}, \mathrm{Ni}$):** The increasing nuclear pull is almost exactly balanced by the growing inter-electronic repulsion among the increasing number of $3d$ electrons. The metallic radii for Fe, Co, and Ni are virtually identical (≈ 125 to $126\,\mathrm{pm}$).
- **Slight Increase at the End ($\mathrm{Cu} \rightarrow \mathrm{Zn}$):** As the $3d$ subshell becomes fully filled ($3d^{10}$), electron-electron repulsion among paired electrons dominates over the nuclear charge, causing a slight expansion of the electron cloud ($\mathrm{Zn} = 137\,\mathrm{pm}$).

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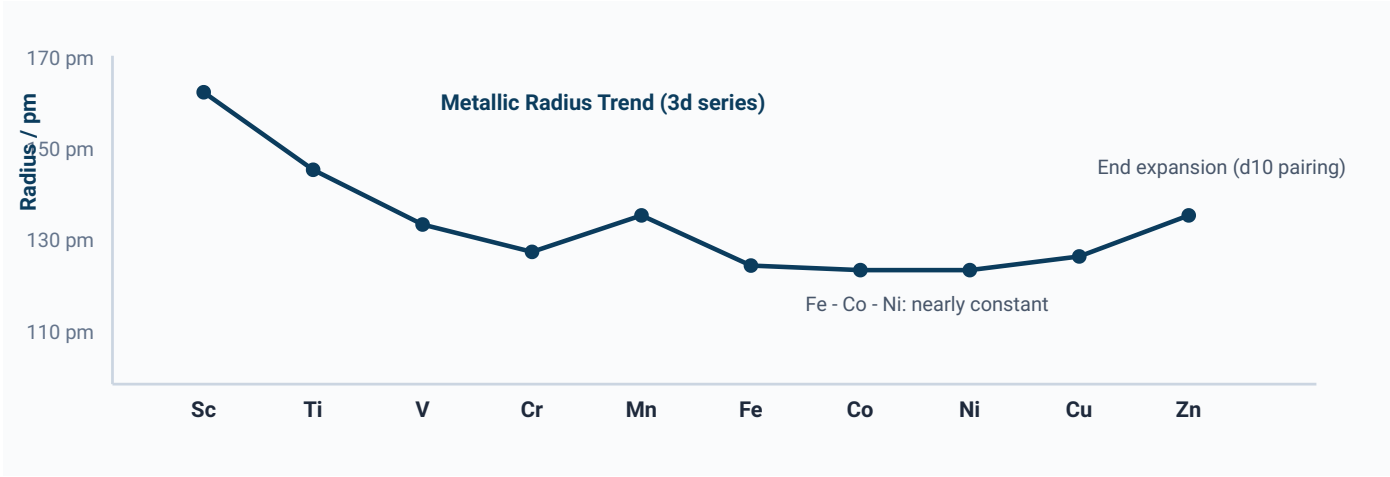


Figure 1: Variation of metallic radii across the 3d series from Scandium to Zinc.

3.2 Enthalpies of Atomization & Melting Points

Transition metals (except Group 12) exhibit exceptionally high melting points and large standard enthalpies of atomization ($\Delta_a H^\circ$). This occurs because electrons from both the outermost ns and the penultimate $(n-1)d$ orbitals participate extensively in metallic bonding.

- **Unpaired Electrons & Bond Strength:** In any transition series, the enthalpy of atomization peaks roughly near the middle (d^5), because a maximal number of unpaired d electrons favors strong interatomic metal-metal interactions.
- **The Manganese and Technetium Anomaly:** Mn ($3d^5 4s^2$) exhibits an abnormally low melting point and a low enthalpy of atomization ($281\ \mathrm{kJ\ mol^{-1}}$). Its stable, half-filled $3d^5$ shell holds electrons tightly, leading to less effective metallic delocalization in its complex crystal lattice.
- **Zinc:** Zinc has the lowest enthalpy of atomization in the entire $3d$ series ($126\ \mathrm{kJ\ mol^{-1}}$) and lowest melting point because no d-electrons participate in metallic bonding (complete $3d^{10}$ shell).
- **Heavy Transition Metals:** Metals of the $4d$ and $5d$ series possess markedly higher enthalpies of atomization than corresponding $3d$ metals. This strong intermetallic bonding explains why metal-metal bonding is far more common in heavy transition elements (e.g., tungsten W, molybdenum Mo).

3.3 Ionization Enthalpies and Exchange Energy

Ionization enthalpies increase along the series, but far less steeply than in s- and p-block elements. This buffering arises because adding electrons to the inner $(n-1)d$ subshell shields the outer $4s$ electrons from the increasing nuclear charge.

- **First Ionization Enthalpy ($\Delta_i H_1$):** Shows an irregular slight increase from Sc to Zn. Zn has an exceptionally high first ionization enthalpy ($906\ \mathrm{kJ\ mol^{-1}}$) because the electron is removed from a stable, completely filled $4s^2$ orbital.
- **Second Ionization Enthalpy ($\Delta_i H_2$):** Cr and Cu exhibit unusually large second ionization enthalpies ($1592\ \mathrm{kJ\ mol^{-1}}$ and $1958\ \mathrm{kJ\ mol^{-1}}$ respectively). Removing a second electron requires disrupting exceptionally stable half-filled $3d^5$ ($\mathrm{Cr^+}$) and fully filled $3d^{10}$ ($\mathrm{Cu^+}$) subshells.
- **Third Ionization Enthalpy ($\Delta_i H_3$):** Shows marked spikes at Mn ($3260\ \mathrm{kJ\ mol^{-1}}$) and Zn ($3837\ \mathrm{kJ\ mol^{-1}}$). Removing an electron from $\mathrm{Mn^{2+}}$ ($3d^5$) destroys its half-filled exchange-stabilized configuration, making $\mathrm{Mn^{3+}}$ energetically demanding to form. Conversely, $\mathrm{Fe^{2+}}$ ($3d^6$) has a lower third ionization enthalpy ($2962\ \mathrm{kJ\ mol^{-1}}$) because electron loss directly achieves the stable $3d^5$ state.

EXCHANGE ENERGY PRINCIPLE

Exchange energy is proportional to the number of possible pairs of parallel spins: $K = \frac{n(n-1)}{2}$. For a d^5 configuration ($n = 5$), there are 10 exchange pairs, providing maximum stabilization. Disruption of this

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state requires compensating with large ionization energy.

4. Oxidation States & Standard Electrode Potentials

4.1 Variable Oxidation States

Transition elements exhibit a wide variety of oxidation states because the energy difference between the inner $(n-1)d$ and outer ns orbitals is very small, allowing electrons from both levels to participate in bond formation.

Table 2: Common Oxidation States of 3d Metals (Bold = Most Common / Stable)

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
+3	+2	+2	+2	**+2**	**+2**	**+2**	**+2**	+1	**+2**
	+3	+3	**+3**	+3	**+3**	**+3**	+3	**+2**	
	+4	+4	+4	+4	+4	+4	+4		
		+5	+5	+5					
			+6	+6	+6				
				+7					

- **Unit Steps vs. Two-Unit Steps:** In transition elements, oxidation states differ by **unity** $(+2, +3, +4, +5)$, reflecting stepwise unpairing and removal of d-electrons. In contrast, p-block oxidation states differ predominantly by **two units** (e.g., $+1$ and $+3$, or $+2$ and $+4$) owing to the inert pair effect.
- **Group Trend Contrasts with p-block:** In the p-block, heavier members prefer lower oxidation states due to the inert pair effect ($\mathrm{Pb}^{2+} > \mathrm{Pb}^{4+}$, $\mathrm{Bi}^{3+} > \mathrm{Bi}^{5+}$). In sharp contrast, within transition metal groups, **heavier members strongly stabilize higher oxidation states**. For example, in Group 6, $\mathrm{Mo(VI)}$ and $\mathrm{W(VI)}$ are exceptionally stable and non-oxidizing, whereas $\mathrm{Cr(VI)}$ in $\mathrm{Cr_2O_7^{2-}}$ is a powerful oxidizing agent.
- **Zero and Low Oxidation States:** Transition metals stabilize zero oxidation states in metal carbonyls such as $[\mathrm{Ni(CO)_4}]$ and $[\mathrm{Fe(CO)_5}]$ via π-acceptor back-bonding ligands.

4.2 Standard Reduction Potentials $E^\circ(\mathrm{M}^{2+}/\mathrm{M})$

The standard reduction potential for the couple $\mathrm{M}^{2+}(\mathrm{aq}) + 2e^- \rightarrow \mathrm{M(s)}$ reflects a balance of three thermochemical terms:

BORN-HABER CYCLE FOR HYDRATED CATION FORMATION

$$\Delta H = \Delta_a H^\circ(\mathrm{M}) + [\Delta_i H_1 + \Delta_i H_2] + \Delta_{\mathrm{hyd}} H^\circ(\mathrm{M}^{2+})$$

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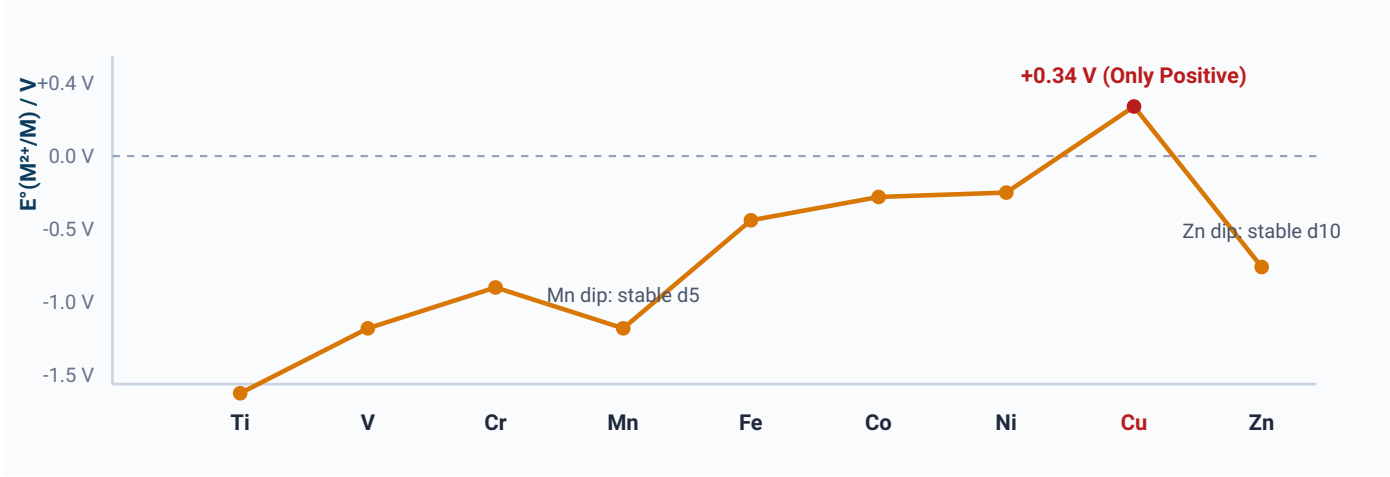


Figure 2: Trends in standard electrode potentials $E^\circ(\mathrm{M}^{2+}/\mathrm{M})$ across 3d transition metals.

- **Why is $E^\circ(\mathrm{Cu}^{2+}/\mathrm{Cu})$ Positive ($+0.34\,\mathrm{V}$)?** Copper has high enthalpy of atomization ($\Delta_a H^\circ = 339\,\mathrm{kJ\,mol^{-1}}$) and very high second ionization enthalpy ($\Delta_i H_2 = 1958\,\mathrm{kJ\,mol^{-1}}$). The energy released during hydration ($\Delta_{\mathrm{hyd}} H^\circ = -2121\,\mathrm{kJ\,mol^{-1}}$) is insufficient to compensate for the sum of these energy inputs. Consequently, Cu cannot reduce H^+ and does not release H_2 from dilute non-oxidizing acids. It dissolves only in oxidizing acids ($\mathrm{HNO_3}$ and hot concentrated $\mathrm{H_2SO_4}$).
- **More Negative Values for Mn, Ni, and Zn:**
  - Mn^{2+} has a more negative E° ($-1.18\,\mathrm{V}$) due to the extra stability of its half-filled d^5 subshell.
  - Zn^{2+} has a more negative E° ($-0.76\,\mathrm{V}$) due to its completely filled d^{10} subshell.
  - Ni^{2+} has a more negative E° ($-0.25\,\mathrm{V}$) due to its exceptionally high negative hydration enthalpy ($-2121\,\mathrm{kJ\,mol^{-1}}$).

4.3 Redox Behavior of the $\mathrm{M}^{3+}/\mathrm{M}^{2+}$ Couple

- **Cr^{2+} is Strongly Reducing vs. Mn^{3+} Strongly Oxidizing:** Both have the d^4 configuration. When Cr^{2+} is oxidized to Cr^{3+}, it attains the d^3 configuration, which possesses a stable, half-filled t_{2g}^3 set in aqueous medium. Conversely, when Mn^{3+} acts as an oxidizing agent and is reduced to Mn^{2+}, it achieves the half-filled, extra-stable $3d^5$ configuration ($E^\circ(\mathrm{Mn}^{3+}/\mathrm{Mn}^{2+}) = +1.57\,\mathrm{V}$).
- **Disproportionation of Copper(I):** $\mathrm{Cu}^+(\mathrm{aq})$ is unstable and disproportionates spontaneously in water:
  $$2\,\mathrm{Cu}^+(\mathrm{aq}) \rightarrow \mathrm{Cu}^{2+}(\mathrm{aq}) + \mathrm{Cu}(\mathrm{s})$$
  Although Cu^+ has a stable $3d^{10}$ configuration, the divalent ion Cu^{2+} has a much higher charge density, yielding a far more negative hydration enthalpy that more than compensates for the second ionization enthalpy of copper.

5. Characteristic Properties of Transition Metal Compounds

5.1 Magnetic Properties and Spin-Only Formula

Paramagnetism results from the presence of unpaired electrons. In the first transition series, the orbital angular momentum is effectively quenched by the surrounding electric field of the ligands. The magnetic moment (μ) is therefore determined strictly by spin:

SPIN-ONLY FORMULA

$$\mu_s = \sqrt{n(n+2)} \quad \text{Bohr Magnetons (BM)}$$

Where n is the number of unpaired electrons. For $n = 1$, $\mu = 1.73\,\mathrm{BM}$; for $n = 5$, $\mu = 5.92\,\mathrm{BM}$.

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5.2 Colour and d–d Transitions

Transition metal ions with partially filled d-orbitals (d^1 to d^9) exhibit distinct colors in aqueous solution. When ligands (like H_2O) approach, degenerate d-orbitals split into different energy levels. Absorption of light in the visible region promotes an electron from a lower to a higher d-orbital (d--d transition). The color observed is the **complementary color** of the absorbed wavelength.

Table 3: Aqueous Colors of First Row Transition Metal Ions

Ion	Configuration	Unpaired Electrons	Solution Color
Sc^{3+}, Ti^{4+}	$3d^0$	0	Colourless (no d-electrons)
Ti^{3+}	$3d^1$	1	Purple
V^{4+} (VO^{2+})	$3d^1$	1	Blue
V^{3+}	$3d^2$	2	Green
V^{2+}	$3d^3$	3	Violet
Cr^{3+}	$3d^3$	3	Violet
Mn^{3+}	$3d^4$	4	Violet
Cr^{2+}	$3d^4$	4	Blue
Mn^{2+}	$3d^5$	5	Pink
Fe^{3+}	$3d^5$	5	Yellow
Fe^{2+}	$3d^6$	4	Green
Co^{2+}	$3d^7$	3	Pink
Ni^{2+}	$3d^8$	2	Green
Cu^{2+}	$3d^9$	1	Blue
Zn^{2+}, Cu^{+}	$3d^{10}$	0	Colourless (completely filled d-shell)

5.3 Catalytic Properties, Interstitial Compounds, and Alloys

- **Catalytic Action:** Transition metals and their oxides act as catalysts due to their ability to exhibit multiple oxidation states, form unstable intermediates, and provide large surface areas with vacant d orbitals. *Classic Example:* Fe^{3+} catalyses the reaction between I^- and peroxodisulphate ($S_2O_8^{2-}$):

$$2\,Fe^{3+} + 2\,I^- \rightarrow 2\,Fe^{2+} + I_2$$
$$2\,Fe^{2+} + S_2O_8^{2-} \rightarrow 2\,Fe^{3+} + 2\,SO_4^{2-}$$

- **Interstitial Compounds:** Formed when small non-metal atoms (H, B, C, N) are trapped inside the interstitial sites of transition metal lattices (e.g., TiC, Mn_4N, Fe_3H, $TiH_{1.7}$, $VH_{0.56}$).
  - They are non-stoichiometric and neither typically ionic nor covalent.
  - They have higher melting points than pure parent metals.
  - They are extremely hard (some borides approach diamond hardness).
  - They retain metallic electrical and thermal conductivity.
  - They are chemically inert.

- **Alloy Formation:** Transition metals have atomic radii within 15% of one another, enabling mutual substitution in crystal lattices to produce hard, high-melting alloys (e.g., brass [Cu--Zn], bronze [Cu--Sn], and stainless steel).

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6. Potassium Dichromate ($K_2Cr_2O_7$) & Potassium Permanganate ($KMnO_4$)

6.1 Potassium Dichromate ($K_2Cr_2O_7$)

Industrial Preparation from Chromite Ore ($FeCr_2O_4$):

1. **Alkaline Fusion:** Chromite ore is fused with sodium carbonate in the presence of air:
$$4\,FeCr_2O_4 + 8\,Na_2CO_3 + 7\,O_2 \rightarrow 8\,Na_2CrO_4\ (\text{yellow}) + 2\,Fe_2O_3 + 8\,CO_2$$
2. **Acidification:** The yellow sodium chromate solution is filtered and acidified with H_2SO_4:
$$2\,Na_2CrO_4 + 2\,H^+ \rightarrow Na_2Cr_2O_7\ (\text{orange}) + 2\,Na^+ + H_2O$$
3. **Potassium Salt Conversion:** Because $Na_2Cr_2O_7$ is highly soluble and deliquescent, it is treated with KCl. $K_2Cr_2O_7$ is less soluble and readily crystallizes as orange crystals:
$$Na_2Cr_2O_7 + 2\,KCl \rightarrow K_2Cr_2O_7 + 2\,NaCl$$

PH DEPENDENCE OF CHROMATE-DICHROMATE EQUILIBRIUM

Chromate (CrO_4^{2-}, yellow) and dichromate ($Cr_2O_7^{2-}$, orange) are interconvertible depending on solution pH. The oxidation state of chromium remains $+6$ in both ions:

$$2\,CrO_4^{2-}\ (\text{yellow}) + 2\,H^+ \rightleftharpoons Cr_2O_7^{2-}\ (\text{orange}) + H_2O$$
$$Cr_2O_7^{2-}\ (\text{orange}) + 2\,OH^- \rightleftharpoons 2\,CrO_4^{2-}\ (\text{yellow}) + H_2O$$

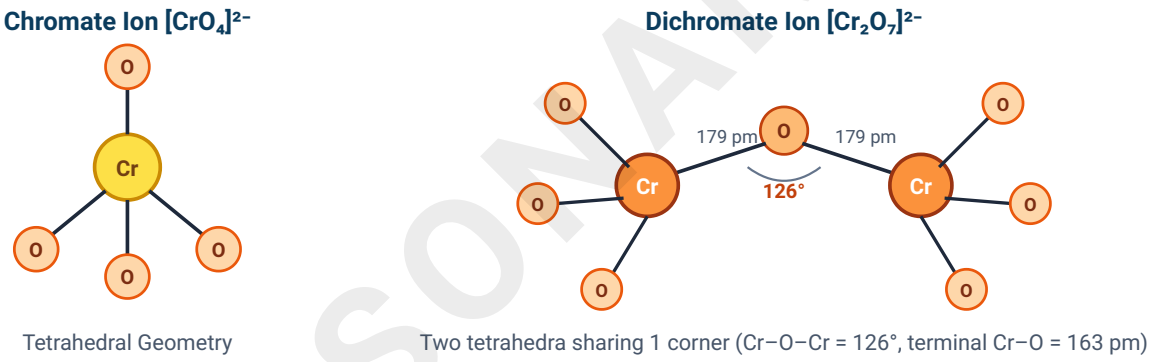


Figure 3: Geometric structures of tetrahedral chromate and corner-shared dichromate ions.

Oxidizing Actions of $K_2Cr_2O_7$ in Acidic Solution ($E^\circ = 1.33\ V$):

STANDARD ACIDIC REDUCTION HALF-REACTION

$$Cr_2O_7^{2-} + 14\,H^+ + 6\,e^- \rightarrow 2\,Cr^{3+} + 7\,H_2O$$

- **With Iodide:** $Cr_2O_7^{2-} + 14\,H^+ + 6\,I^- \rightarrow 2\,Cr^{3+} + 3\,I_2 + 7\,H_2O$
- **With Iron(II):** $Cr_2O_7^{2-} + 14\,H^+ + 6\,Fe^{2+} \rightarrow 2\,Cr^{3+} + 6\,Fe^{3+} + 7\,H_2O$
- **With H_2S:** $Cr_2O_7^{2-} + 8\,H^+ + 3\,H_2S \rightarrow 2\,Cr^{3+} + 3\,S\ (\text{precipitate}) + 7\,H_2O$
- **With Tin(II):** $Cr_2O_7^{2-} + 14\,H^+ + 3\,Sn^{2+} \rightarrow 2\,Cr^{3+} + 3\,Sn^{4+} + 7\,H_2O$

6.2 Potassium Permanganate ($KMnO_4$)

Preparation:

- **Commercial Preparation:** Pyrolusite ore (MnO_2) undergoes alkaline oxidative fusion with KOH and O_2 (or KNO_3) to yield dark green potassium manganate:
$$2\,MnO_2 + 4\,KOH + O_2 \rightarrow 2\,K_2MnO_4\ (\text{dark green}) + 2\,H_2O$$
The manganate is then oxidized to permanganate electrolytically in alkaline solution:

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$$\mathrm{MnO_4^{2-}\ (green)} \xrightarrow{\text{electrolytic oxidation}} \mathrm{MnO_4^-\ (purple)} + e^-$$

- **Disproportionation Route:** In neutral or acidic media, manganate disproportionates spontaneously:
$$3\,\mathrm{MnO_4^{2-}} + 4\,\mathrm{H^+} \rightarrow 2\,\mathrm{MnO_4^-} + \mathrm{MnO_2} + 2\,\mathrm{H_2O}$$
- **Laboratory Preparation:** Direct oxidation of $\mathrm{Mn^{2+}}$ salt with peroxodisulphate:
$$2\,\mathrm{Mn^{2+}} + 5\,\mathrm{S_2O_8^{2-}} + 8\,\mathrm{H_2O} \rightarrow 2\,\mathrm{MnO_4^-} + 10\,\mathrm{SO_4^{2-}} + 16\,\mathrm{H^+}$$



Figure 4: Tetrahedral structures, oxidation states, and magnetic properties of manganate and permanganate.

HIGH-YIELD NEET FACT: ORIGIN OF COLOUR IN PERMANGANATE

In $\mathrm{MnO_4^-}$, manganese is in the $+7$ state with a completely empty d-subshell (d^0). Therefore, its intense purple color **cannot arise from d--d transitions**. Instead, it originates from **Ligand-to-Metal Charge Transfer (LMCT)**, where electron density temporarily transfers from oxygen p-orbitals to empty manganese d-orbitals upon photon absorption.

6.3 Reactions of $\mathrm{KMnO_4}$ Across Different Media

- **1. In Strongly Acidic Medium ($\mathrm{H^+}$, $5e^-$ transfer, $E^\circ = +1.52\ \mathrm{V}$):**
$$\mathrm{MnO_4^-} + 8\,\mathrm{H^+} + 5\,e^- \rightarrow \mathrm{Mn^{2+}} + 4\,\mathrm{H_2O}$$
  - **Oxalate / Oxalic acid:** $5\,\mathrm{C_2O_4^{2-}} + 2\,\mathrm{MnO_4^-} + 16\,\mathrm{H^+} \rightarrow 2\,\mathrm{Mn^{2+}} + 10\,\mathrm{CO_2} + 8\,\mathrm{H_2O}$ (carried out at $333\ \mathrm{K}$)
  - **Iron(II):** $5\,\mathrm{Fe^{2+}} + \mathrm{MnO_4^-} + 8\,\mathrm{H^+} \rightarrow \mathrm{Mn^{2+}} + 5\,\mathrm{Fe^{3+}} + 4\,\mathrm{H_2O}$
  - **Nitrite:** $5\,\mathrm{NO_2^-} + 2\,\mathrm{MnO_4^-} + 6\,\mathrm{H^+} \rightarrow 2\,\mathrm{Mn^{2+}} + 5\,\mathrm{NO_3^-} + 3\,\mathrm{H_2O}$
  - **Iodide:** $10\,\mathrm{I^-} + 2\,\mathrm{MnO_4^-} + 16\,\mathrm{H^+} \rightarrow 2\,\mathrm{Mn^{2+}} + 5\,\mathrm{I_2} + 8\,\mathrm{H_2O}$
  - **Sulphite:** $5\,\mathrm{SO_3^{2-}} + 2\,\mathrm{MnO_4^-} + 6\,\mathrm{H^+} \rightarrow 2\,\mathrm{Mn^{2+}} + 5\,\mathrm{SO_4^{2-}} + 3\,\mathrm{H_2O}$
- **2. In Neutral or Faintly Alkaline Medium ($3e^-$ transfer, $E^\circ = +1.69\ \mathrm{V}$):**
$$\mathrm{MnO_4^-} + 2\,\mathrm{H_2O} + 3\,e^- \rightarrow \mathrm{MnO_2} + 4\,\mathrm{OH^-}$$
  - **Iodide to Iodate:** $2\,\mathrm{MnO_4^-} + \mathrm{H_2O} + \mathrm{I^-} \rightarrow 2\,\mathrm{MnO_2} + 2\,\mathrm{OH^-} + \mathrm{IO_3^-}$ (Contrast with acidic medium, which yields $\mathrm{I_2}$!)
  - **Thiosulphate to Sulphate:** $8\,\mathrm{MnO_4^-} + 3\,\mathrm{S_2O_3^{2-}} + \mathrm{H_2O} \rightarrow 8\,\mathrm{MnO_2} + 6\,\mathrm{SO_4^{2-}} + 2\,\mathrm{OH^-}$
  - **Manganous Salt to $\mathrm{MnO_2}$:** $2\,\mathrm{MnO_4^-} + 3\,\mathrm{Mn^{2+}} + 2\,\mathrm{H_2O} \rightarrow 5\,\mathrm{MnO_2} + 4\,\mathrm{H^+}$

NEET ANALYTICAL WARNING: TITRATIONS WITH HCL

Permanganate titrations cannot be conducted in the presence of HCl because $\mathrm{KMnO_4}$ oxidizes hydrochloric acid to chlorine gas ($2\,\mathrm{MnO_4^-} + 16\,\mathrm{H^+} + 10\,\mathrm{Cl^-} \rightarrow 2\,\mathrm{Mn^{2+}} + 5\,\mathrm{Cl_2} + 8\,\mathrm{H_2O}$), consuming titrant errantly. Dilute $\mathrm{H_2SO_4}$ is the only suitable acid.

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7. The Inner Transition Elements (f-Block)

7.1 The Lanthanoids (4f Series, Ce to Lu)

Lanthanoids comprise the 14 elements following Lanthanum (La, $Z = 57$). Their general electronic configuration is $[\mathrm{Xe}]\,4f^{1-14}\,5d^{0-1}\,6s^2$. For all tripositive ions (Ln^{3+}), the configuration simplifies strictly to $[\mathrm{Xe}]\,4f^n$ ($n = 0$ to 14).

Table 4: Electronic Configurations and Radii of Lanthanum and Lanthanoids

Z	Element	Symbol	Outer Configuration (Ln)	Ln^{3+}	Atomic Radius (pm)	Ln^{3+} Radius (pm)
57	Lanthanum	La	$5d^1 6s^2$	$4f^0$	187	106
58	Cerium	Ce	$4f^1 5d^1 6s^2$	$4f^1$	183	103
59	Praseodymium	Pr	$4f^3 6s^2$	$4f^2$	182	101
60	Neodymium	Nd	$4f^4 6s^2$	$4f^3$	181	99
61	Promethium	Pm	$4f^5 6s^2$	$4f^4$	181	98
62	Samarium	Sm	$4f^6 6s^2$	$4f^5$	180	96
63	Europium	Eu	$4f^7 6s^2$	$4f^6$	199	95
64	Gadolinium	Gd	$4f^7 5d^1 6s^2$	$4f^7$	180	94
65	Terbium	Tb	$4f^9 6s^2$	$4f^8$	178	92
66	Dysprosium	Dy	$4f^{10} 6s^2$	$4f^9$	177	91
67	Holmium	Ho	$4f^{11} 6s^2$	$4f^{10}$	176	89
68	Erbium	Er	$4f^{12} 6s^2$	$4f^{11}$	175	88
69	Thulium	Tm	$4f^{13} 6s^2$	$4f^{12}$	174	87
70	Ytterbium	Yb	$4f^{14} 6s^2$	$4f^{13}$	173	86
71	Lutetium	Lu	$4f^{14} 5d^1 6s^2$	$4f^{14}$	–	–

7.2 Lanthanoid Contraction and Its Consequences

As atomic number increases across the lanthanoid series, each added electron enters the inner $4f$ subshell. The diffused, complex shape of $4f$ orbitals results in **extremely poor shielding** of one $4f$ electron by another. Consequently, the increasing nuclear charge draws the outer shells inward, producing a steady, regular decrease in both atomic and Ln^{3+} ionic radii (from $106\ \mathrm{pm}$ in La^{3+} to $86\ \mathrm{pm}$ in Yb^{3+}).

THREE MAJOR NEET CONSEQUENCES OF LANTHANOID CONTRACTION

1. **Identical Radii of 4d and 5d Elements:** The contraction in the intervening $4f$ series completely compensates for the normal radius increase down a group. Hence, pairs like Zr **($160\ \mathrm{pm}$) and** Hf **(** $159\ \mathrm{pm}$**)**, and Nb and Ta, possess virtually identical radii and identical chemical properties, making separation difficult.
2. **Basic Strength of Hydroxides Decreases:** From $\mathrm{La(OH)_3}$ to $\mathrm{Lu(OH)_3}$, the size of Ln^{3+} decreases. By Fajan's rules, covalent character of the $\mathrm{Ln\text{--}OH}$ bond increases and ionic character decreases. Thus, $\mathrm{La(OH)_3}$ **is the most basic** and $\mathrm{Lu(OH)_3}$ **is the least basic**.
3. **Separation by Ion-Exchange:** Small, regular size variations allow subtle differences in complex formation stability, enabling separation via ion-exchange resins.

7.3 Oxidation States and Redox Peculiarities of Lanthanoids

The predominant and most stable oxidation state of all lanthanoids is **+3**. However, $+2$ and $+4$ states arise due to the stability of empty (f^0), half-filled (f^7), or fully filled (f^{14}) subshells:

BOSONANCE

- Ce^{4+} **($4f^0$):** Possesses noble-gas configuration. It is an excellent analytical oxidizing agent ($E^\circ(\mathrm{Ce}^{4+}/\mathrm{Ce}^{3+}) = +1.74\,\mathrm{V}$), reverting readily to the favored $+3$ state.
- Tb^{4+} **($4f^7$):** Achieves a half-filled $4f^7$ shell; acts as an oxidizing agent.
- Eu^{2+} **($4f^7$) and** Yb^{2+} **($4f^{14}$):** Eu^{2+} is formed by losing two $6s$ electrons, leaving a stable half-filled $4f^7$ subshell. It acts as a powerful **reducing agent** to return to the favored $+3$ state. Similarly, Yb^{2+} ($4f^{14}$) is a strong reductant.

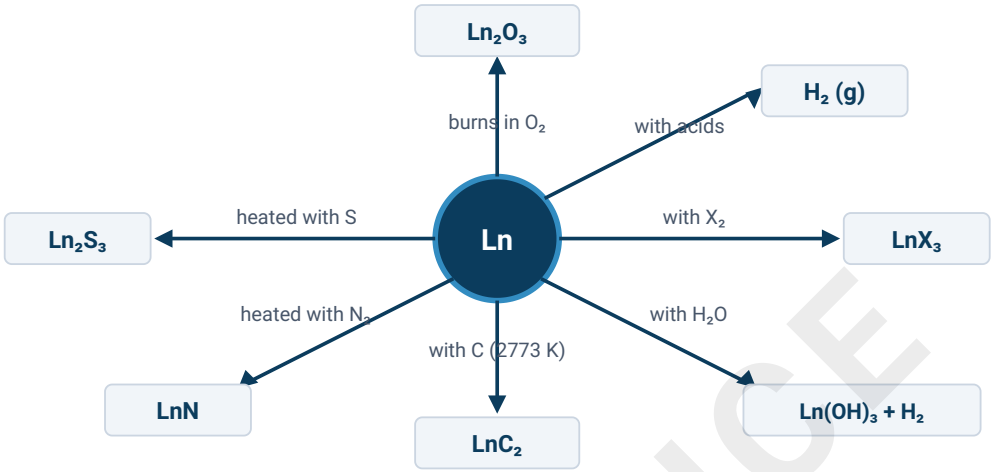


Figure 5: Summary of fundamental chemical reactions of lanthanoid metals.

Industrial Application — Mischmetall: A commercially vital pyrophoric alloy containing $\sim 95\%$ lanthanoid metals (mainly Ce, La, Nd), $\sim 5\%$ iron, and traces of S, C, Ca, and Al. Used in magnesium-based alloys to manufacture lighter flints, bullet jackets, and artillery shells.

7.4 The Actinoids (5f Series, Th to Lr)

The actinoids comprise 14 radioactive elements from Thorium ($Z = 90$) to Lawrencium ($Z = 103$). Their general electronic configuration is $[\mathrm{Rn}]\,5f^{1-14}\,6d^{0-1}\,7s^2$.

- **Wide Range of Oxidation States:** Unlike lanthanoids, actinoids exhibit a much wider range of oxidation states (up to $+7$ in Np and Pu). This happens because the energy gap between $5f$, $6d$, and $7s$ orbitals is very small, allowing electrons from all three levels to participate in bonding.
- **Actinoid Contraction:** There is a gradual decrease in atomic and tripositive ionic radii across the series. The **actinoid contraction is significantly greater from element to element than the lanthanoid contraction** because $5f$ electrons have even poorer shielding ability than $4f$ electrons.
- **Chemical Reactivity:** Actinoids are far more reactive than lanthanoids due to lower ionization enthalpies (their $5f$ electrons are less deeply buried than $4f$ electrons and participate more readily in bond formation).

Table 5: Comprehensive Comparison: Lanthanoids vs. Actinoids

Characteristic	Lanthanoids (4f Series)	Actinoids (5f Series)
Differentiating Subshell	Electrons enter the $4f$ subshell.	Electrons enter the $5f$ subshell.
Oxidation States	Exhibit primarily $+3$; rarely $+2$ or $+4$.	Exhibit a wide spectrum ($+3, +4, +5, +6, +7$).
Energy Levels	Large energy gap between $4f$ and $5d$.	Comparable energy between $5f$, $6d$, and $7s$.
Contraction Magnitude	Smaller contraction per element.	Greater contraction per element (poorer $5f$ shielding).
Chemical Reactivity	Less reactive (similar to calcium/aluminium).	Highly reactive, especially when finely divided.
Complex Formation	Lower tendency to form complexes.	High tendency to form complexes.

BOSONANCE

Characteristic	Lanthanoids (4f Series)	Actinoids (5f Series)
Basic Strength of Hydroxides	Less basic hydroxides.	More basic hydroxides.
Radioactivity	Non-radioactive (except Promethium, Pm).	All are radioactive.
Magnetic Properties	Simpler; spin and orbital contributions easily separable.	More complex due to extensive $5f$ delocalization.

8. Worked Conceptual Examples

Worked Example 1: Spin-Only Magnetic Moment Calculation

NEET Core Calculation

Problem: Calculate the spin-only magnetic moment of a divalent metal ion in aqueous solution if its atomic number is $Z = 27$.

Solution:

1. Neutral atom configuration ($Z = 27$, Cobalt): $[\mathrm{Ar}]\, 3d^7 4s^2$.
2. Ionization to form divalent cation (Co^{2+}): The two $4s$ electrons are removed first. Configuration = $[\mathrm{Ar}]\, 3d^7$.
3. Determine unpaired electrons: Five $3d$ orbitals accommodate 7 electrons. According to Hund's rule, two orbitals contain paired electrons and three orbitals contain single unpaired electrons. Thus, $n = 3$.
4. Calculate magnetic moment:

$$\mu_s = \sqrt{n(n+2)} = \sqrt{3(3+2)} = \sqrt{15} \approx 3.87\,\mathrm{BM}$$

Worked Example 2: Balancing Acidic Permanganate Redox Equations

Stoichiometric Reasoning

Problem: Write and balance the complete ionic equation for the oxidation of oxalate ion ($\mathrm{C_2O_4^{2-}}$) by acidified potassium permanganate solution.

Solution:

1. Reduction half-reaction: Permanganate in acidic medium gains 5 electrons:

$$\mathrm{MnO_4^-} + 8\,\mathrm{H^+} + 5\,e^- \rightarrow \mathrm{Mn^{2+}} + 4\,\mathrm{H_2O} \quad (\times 2)$$

2. Oxidation half-reaction: Oxalate loses 2 electrons to form $\mathrm{CO_2}$:

$$\mathrm{C_2O_4^{2-}} \rightarrow 2\,\mathrm{CO_2} + 2\,e^- \quad (\times 5)$$

3. Multiply to equalize electrons ($10\,e^-$ each) and combine:

$$2\,\mathrm{MnO_4^-} + 5\,\mathrm{C_2O_4^{2-}} + 16\,\mathrm{H^+} \rightarrow 2\,\mathrm{Mn^{2+}} + 10\,\mathrm{CO_2} + 8\,\mathrm{H_2O}$$

BOSONANCE

Worked Example 3: Lanthanoid Contraction and Chemical Similarity

Periodic Anomaly Analysis

Problem: Why do Zirconium (Zr, $Z = 40$) and Hafnium (Hf, $Z = 72$) exhibit almost identical physical and chemical properties and occur together in minerals, despite belonging to different periods?

Solution:

Zirconium belongs to the second transition series ($4d$) and Hafnium belongs to the third transition series ($5d$). Normally, moving from period 5 to period 6 increases atomic radius due to the addition of an extra electron shell. However, intervening between Lanthanum ($Z = 57$) and Hafnium ($Z = 72$) are the 14 lanthanoid elements, in which the $4f$ subshell is progressively filled.

Because $4f$ electrons provide extremely poor shielding, the steady increase in nuclear charge pulls the outer valence shell inward (lanthanoid contraction). This contraction almost perfectly counterbalances the normal size increase down the group. As a result, the atomic radius of Zr is $160\ \mathrm{pm}$ and that of Hf is $159\ \mathrm{pm}$. Their identical sizes and identical valence configurations result in nearly indistinguishable chemical properties.

9. Original NEET UG Practice Questions (With Detailed Solutions)

Practice Question 1 (Single Correct MCQ)

Topic: Electronic Configurations & Definitions

Which one of the following statements is **incorrect** regarding transition and group 12 elements?

(A) Zn, Cd, and Hg are not regarded as transition metals because they have full d^{10} configurations in their ground states as well as common oxidation states.

(B) Silver is regarded as a transition element because it can exhibit an incompletely filled $4d$ orbital in its $+2$ oxidation state.

(C) Scandium ($Z = 21$) is not a transition element because its $+3$ ion has a $3d^0$ configuration.

(D) Copper is a transition element because its divalent ion (Cu^{2+}) has a $3d^9$ configuration.

Correct Answer: (C)

Worked Reasoning: By IUPAC definition, an element is classified as a transition metal if it has an incompletely filled d-subshell in its neutral atom OR in any of its common ions. Scandium in its elemental ground state has the configuration $[\mathrm{Ar}]\,3d^1 4s^2$. Because its neutral atom contains an unfilled $3d$ orbital ($3d^1$), Scandium **is** definitively a transition element. Therefore, statement (C) is false.

BOSONANCE

Practice Question 2 (Single Correct MCQ)

Topic: Thermochemical Trends & Redox Potentials

The standard reduction potential $E^\circ(\mathrm{M}^{2+}/\mathrm{M})$ is positive for copper ($+0.34\,\mathrm{V}$). The principal thermochemical reason for this behavior is that:

(A) Copper has an exceptionally low enthalpy of atomization.

(B) The high enthalpy of atomization and high second ionization enthalpy of copper are not balanced by its hydration enthalpy.

(C) The hydration enthalpy of Cu^{2+} is the lowest among all $3d$-series divalent ions.

(D) Copper easily loses two electrons to achieve an empty $4s$ orbital.

Correct Answer: (B)

Worked Reasoning: The standard reduction potential $E^\circ(\mathrm{M}^{2+}/\mathrm{M})$ depends on the sum of atomization enthalpy ($\Delta_a H^\circ$), first and second ionization enthalpies ($\Delta_i H_1 + \Delta_i H_2$), and hydration enthalpy ($\Delta_{\mathrm{hyd}} H^\circ$). For copper, the sum of $\Delta_a H^\circ$ ($339\,\mathrm{kJ\,mol^{-1}}$) and the exceptionally high $\Delta_i H_2$ ($1958\,\mathrm{kJ\,mol^{-1}}$, required to break the stable $3d^{10}$ shell) is so large that even the substantial hydration enthalpy of Cu^{2+} ($-2121\,\mathrm{kJ\,mol^{-1}}$) cannot compensate for it. Hence, the overall process is endothermic and E° is positive.

Practice Question 3 (Assertion-Reasoning)

Topic: Reducing/Oxidizing Strength of d4 Ions

Given below are two statements labeled as **Assertion (A)** and **Reason (R)**:

Assertion (A): Cr^{2+} is a strong reducing agent in aqueous solution, whereas Mn^{3+} is a strong oxidizing agent, even though both have a d^4 electronic configuration.

Reason (R): Oxidation of Cr^{2+} results in a stable half-filled t_{2g}^3 crystal field configuration in water, while reduction of Mn^{3+} produces the half-filled d^5 configuration.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Answer: (A)

Worked Reasoning: In aqueous solution, octahedral coordination splits d-orbitals into t_{2g} and e_g sets. When Cr^{2+} (d^4) loses an electron to become Cr^{3+} (d^3), it achieves a half-filled t_{2g}^3 configuration of enhanced stability. For Mn^{3+} (d^4), gaining an electron gives Mn^{2+} (d^5), achieving extra stability from a half-filled d-subshell. Both statements are true and Reason (R) directly explains Assertion (A).

BOSONANCE

Practice Question 4 (Multi-Statement Evaluation) Topic: Potassium Permanganate Chemistry

Consider the following four statements regarding $\mathrm{KMnO_4}$ and identify the **correct** combination:

1. In neutral or faintly alkaline medium, $\mathrm{KMnO_4}$ oxidizes iodide ($\mathrm{I^-}$) to iodate ($\mathrm{IO_3^-}$).
2. Permanganate ion ($\mathrm{MnO_4^-}$) is intense purple in colour due to d--d electron transitions.
3. Green manganate ion ($\mathrm{MnO_4^{2-}}$) is paramagnetic with one unpaired electron, while purple permanganate ion ($\mathrm{MnO_4^-}$) is diamagnetic.
4. Hydrochloric acid can be used safely to acidify $\mathrm{KMnO_4}$ in quantitative volumetric titrations.

(A) 1 and 2 only

(B) 1 and 3 only

(C) 2, 3, and 4 only

(D) 1, 3, and 4 only

Correct Answer: (B)

Worked Reasoning:

• Statement 1 is **correct**: In neutral or alkaline medium, $\mathrm{I^-}$ is oxidized to $\mathrm{IO_3^-}$ (in acidic medium, it is oxidized to $\mathrm{I_2}$).

• Statement 2 is **incorrect**: In $\mathrm{MnO_4^-}$, manganese is in $+7$ state ($3d^0$). The intense colour is caused by Ligand-to-Metal Charge Transfer (LMCT), not d--d transitions.

• Statement 3 is **correct**: $\mathrm{MnO_4^{2-}}$ has $\mathrm{Mn(VI)}$ ($3d^1$, 1 unpaired electron $\rightarrow$ paramagnetic), while $\mathrm{MnO_4^-}$ has $\mathrm{Mn(VII)}$ ($3d^0$, diamagnetic).

• Statement 4 is **incorrect**: HCl cannot be used because $\mathrm{KMnO_4}$ oxidizes $\mathrm{Cl^-}$ to $\mathrm{Cl_2}$ gas, which spoils titration accuracy.

BOSONANCE

Practice Question 5 (Matching Column Grid)

Topic: Industrial Catalysts and Applications

Match the catalyst/species listed in **Column I** with its specific industrial application in **Column II**:

Column I (Catalyst / Reagent)	Column II (Industrial Process)
(A) V_2O_5	1. Manufacture of polyethylene (polythene)
(B) $TiCl_4 + Al(CH_3)_3$	2. Oxidation of SO_2 to SO_3 in Contact Process
(C) $PdCl_2$	3. Haber process for ammonia synthesis
(D) Finely divided Fe	4. Wacker process for oxidation of ethyne to ethanal

(A) A-2, B-1, C-4, D-3

(B) A-1, B-2, C-3, D-4

(C) A-2, B-4, C-1, D-3

(D) A-3, B-1, C-4, D-2

Correct Answer: (A)

Worked Reasoning:
- V_2O_5 catalyzes $SO_2 \rightarrow SO_3$ in the Contact Process for H_2SO_4 manufacture (A $\rightarrow$ 2).
- $TiCl_4 + Al(CH_3)_3$ is the Ziegler-Natta catalyst used in high-density polyethylene synthesis (B $\rightarrow$ 1).
- $PdCl_2$ is the catalyst in the Wacker process for converting ethyne/ethene to ethanal (C $\rightarrow$ 4).
- Finely divided Fe with promoters catalyzes the Haber process for NH_3 (D $\rightarrow$ 3).

Practice Question 6 (Single Correct MCQ)

Topic: Lanthanoid vs. Actinoid Contraction

Actinoid contraction is greater from element to element than lanthanoid contraction. What is the fundamental physical reason for this difference?

(A) $5f$ electrons have greater shielding effect than $4f$ electrons.

(B) $5f$ electrons penetrate deeper into the inner core than $4f$ electrons.

(C) The shielding of nuclear charge by $5f$ electrons is poorer than that by $4f$ electrons.

(D) Actinoids have smaller atomic masses than lanthanoids.

Correct Answer: (C)

Worked Reasoning: Because $5f$ orbitals are more diffuse and extend further into space than $4f$ orbitals, their spatial distribution makes them even less effective at shielding each other from the increasing nuclear charge. As atomic number increases, the net electrostatic attraction exerted by the nucleus on outer electrons rises more steeply per element in actinoids than in lanthanoids, yielding a greater contraction.

BOSONANCE

Practice Question 7 (Single Correct MCQ)

Topic: Potassium Dichromate Structure & Bonding

In the dichromate dianion ($Cr_2O_7^{2-}$), which of the following geometric and bonding descriptions is strictly accurate?

(A) It consists of two octahedra sharing an edge with a $Cr\text{--}O\text{--}Cr$ angle of 180°.

(B) It consists of two tetrahedra sharing one corner with a $Cr\text{--}O\text{--}Cr$ bond angle of 126°.

(C) All eight $Cr\text{--}O$ bond lengths are strictly identical at $179\ pm$.

(D) The oxidation state of chromium is $+3$ in chromate and $+6$ in dichromate.

Correct Answer: (B)

Worked Reasoning: The dichromate ion consists of two tetrahedral CrO_4 units sharing one common bridging oxygen atom. The $Cr\text{--}O\text{--}Cr$ bond angle is 126°. The bridging $Cr\text{--}O$ bonds are longer ($179\ pm$), whereas the six terminal $Cr\text{--}O$ bonds are shorter ($163\ pm$) and equivalent due to resonance. In both chromate and dichromate, chromium is in the $+6$ oxidation state.

Practice Question 8 (Single Correct MCQ)

Topic: Interstitial Compounds

Which of the following is **not** a characteristic property of interstitial compounds formed by transition metals?

(A) They have high melting points, higher than those of the pure parent metals.

(B) They are extremely hard, with some borides approaching diamond in hardness.

(C) They completely lose metallic electrical conductivity and become electrical insulators.

(D) They are chemically inert and non-stoichiometric in composition.

Correct Answer: (C)

Worked Reasoning: Interstitial compounds **retain metallic electrical and thermal conductivity** because the metallic bonding network of the host lattice remains intact while small atoms (H, C, N, B) occupy vacant interstitial voids. Therefore, statement (C) is false.

10. Common Student Pitfalls & High-Yield NEET Checklist

⚠️ Pitfall 1: Oxidation State Discrepancy

Students often mistake the change in oxidation number when $KMnO_4$ reacts in different media. Remember:

⚠️ Pitfall 2: Permanganate with Iodide

Pay strict attention to the reaction medium specified in the question:

- In **acidic medium**, MnO_4^- oxidizes I^- to **free iodine (I_2)**.

BOSONANCE

- **Acidic:** $\mathrm{Mn(VII)} \rightarrow \mathrm{Mn^{2+}}$ (Change of $5e^-$, equivalent weight = $M/5$)
- **Neutral / Faintly Alkaline:** $\mathrm{Mn(VII)} \rightarrow \mathrm{MnO_2}$ (Change of $3e^-$, equivalent weight = $M/3$)
- **Strongly Alkaline:** $\mathrm{Mn(VII)} \rightarrow \mathrm{MnO_4^{2-}}$ (Change of $1e^-$, equivalent weight = $M/1$)

- In **faintly alkaline / neutral medium**, $\mathrm{MnO_4^-}$ oxidizes $\mathrm{I^-}$ to **iodate** ($\mathrm{IO_3^-}$).

⚠️ Pitfall 3: Basic Strength of Lanthanoids

Do not reverse the hydroxide trend:

- $\mathrm{La^{3+}}$ is largest $\rightarrow$ least covalent $\rightarrow$ $\mathrm{La(OH)_3}$ **is most basic**.
- $\mathrm{Lu^{3+}}$ is smallest $\rightarrow$ most covalent $\rightarrow$ $\mathrm{Lu(OH)_3}$ **is least basic**.

⚠️ Pitfall 4: Group Trends in Stability

Never apply the p-block inert pair effect to d-block transition metals:

- In p-block: Lower oxidation states are more stable down the group ($\mathrm{Pb^{2+}} > \mathrm{Pb^{4+}}$).
- In d-block: **Higher oxidation states are more stable down the group** ($\mathrm{W(VI)} > \mathrm{Mo(VI)} \gg \mathrm{Cr(VI)}$).

11. Concise Final Recap

KEY FORMULAE & FACTS FOR NEET UG

- **Spin-Only Magnetic Moment:** $\mu_s = \sqrt{n(n+2)}\,\mathrm{BM}$. If $\mu \approx 1.73 \rightarrow n = 1$; $\mu \approx 2.84 \rightarrow n = 2$; $\mu \approx 3.87 \rightarrow n = 3$; $\mu \approx 4.90 \rightarrow n = 4$; $\mu \approx 5.92 \rightarrow n = 5$.
- **Why Copper is Positive:** $\Delta_a H^\circ + \Delta_i H_1 + \Delta_i H_2 > |\Delta_{\mathrm{hyd}} H^\circ|$. Cannot liberate $\mathrm{H_2}$ from dilute non-oxidizing acids.
- **Stability of $\mathrm{Cu^{2+}}$ over $\mathrm{Cu^+}$:** High hydration energy of divalent $\mathrm{Cu^{2+}}$ drives the disproportionation $2\,\mathrm{Cu^+}(aq) \rightarrow \mathrm{Cu^{2+}}(aq) + \mathrm{Cu}(s)$.
- **Chromate-Dichromate Conversion:** $2\,\mathrm{CrO_4^{2-}} + 2\,\mathrm{H^+} \rightleftharpoons \mathrm{Cr_2O_7^{2-}} + \mathrm{H_2O}$. In dichromate, bridging Cr--O--Cr is 126° (179 pm), and terminal Cr--O is 163 pm.
- **Lanthanoid Contraction:** Inefficient shielding of $4f$ electrons causes Zr (160 pm) and Hf (159 pm) to have virtually identical radii.
- **Actinoid Contraction:** Greater than lanthanoid contraction due to poorer shielding by $5f$ electrons. Actinoids exhibit wider oxidation states ($+3$ to $+7$) because $5f$, $6d$, and $7s$ are close in energy.

BOSONANCE